ITaCaS: a serious game and an expert system to support the teaching of sexual-health and hygiene for youth with intellectual disability

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***Abstract*—In Ecuador, according to the National Council for Disability Equality (CONADIS, acronym in Spanish) there are 485,325 people with disabilities, being the intellectual disability the one with the highest incidence, with a prevalence of 108,312 people. In developing countries, there are still insufficient resources for teaching Sexual Health and Hygiene (SHH) to people with disabilities. Thus, it is essential to apply strategies that help teenagers with intellectual disabilities to learn about sexual health and self-care. Therefore, this article presents an educational game and an expert system called "I take care of myself" (ITaCaS) which aims to assist in the teaching of the SHH for young people with intellectual disabilities, with ages between 12 and 17 years old. The expert system helps to determine educational activities to be carried out by the teacher with the teenager, based on their profile and prior knowledge of 13 SHH educational topics. The pictures of the game were specially designed for disabled people. The proposal was validated through a study that was carried out in two special education centers with 30 experts in the field of special education. The obtained results are encouraging since they show great acceptance by the participants of this experiment.**

I. Introduction

Intellectual disability is a life condition characterized by limitations in the intellectual functioning and in adaptive behaviors. The person with an intellectual disability maintains a slow learning and reasoning rhythm [1], furthermore, he will also present communication difficulties, aggressive behaviors and sexual disinhibition. This last feature causes people with this type of disability to be easily influenced by the environment.

Therefore, it is essential to develop tools to support the teaching of Sexual Health and Hygiene (SHH) for people with this disability, since it has been stigmatized by society, many myths and false beliefs have been created, as well as family overprotection and infantilization. The society does not recognize the sexual development of these people, affecting their possibilities of social interaction and development. In addition, parents and teachers perceive as "abnormal" the sexual manifestations and needs of teenagers with intellectual disabilities, causing them to repress their sexual impulse. Consequently, over time, their emotional state will be negatively affected, causing anxiety and distress crises [2].

In this line, it is important to highlight that intellectual disability affects other areas such as behavioral, it prevents them to set above the consequences of their actions into the environment and into their emotional area, since they present an inability to control their emotions and moods [3]. Hence, it must be taken into account that people with intellectual disabilities require educational support in the sexual affective relationship that allows them to regulate and improve their masturbation behaviors in public spaces and to prevent them from being abused, since this population is considered to be the most vulnerable to the risk of sexual abuse [4].

In the specific case of Ecuador, the Constitution indicates in the art. 32 that "*health is a right guaranteed by the State, which is linked to the exercise of other rights, (...). The State shall guarantee this right by means of economic, social, cultural, educational and environmental policies; and the permanent, timely and non-exclusive access to programs, actions and services promoting and providing integral healthcare, sexual health and reproductive health*" [5].

Despite current regulations, in Ecuador there are deficiencies in educational resources and recreational tools on self-care and sexuality for teenagers with intellectual disabilities. Existing resources are scarce and are not adapted for peo-

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ple with intellectual disabilities. Furthermore, learning the contents of the aforementioned resources is complex, since they focus on abstract and not very illustrative messages, generating lack of understanding or disinformation. Therefore, this article presents an educational game and an expert system that relies on rule-based reasoning in order to provide support in teaching SHH to young people with intellectual disabilities and between 12 and 17 years old. The game is organized in three large sections: Changes in the body of men and women, Myths about sexuality and Sexual abuse prevention. On the other hand, the expert system suggests what educational strategies should be applied by teachers according to the level of intellectual disability (mild or moderate) and to the prior knowledge that the teenager has about 13 topics of sexual health, hygiene and abuse-prevention.

The rest of the paper is organized as follows. In Section 2, we present some relevant proposals related to Sexual and Reproductive Health. The system's general architecture as well as the expert system, are described in Section 3. In Section 4, we present the results obtained by the expert system and the perceptions of 30 experts in the special education area. Conclusion and future work are described in Section 5.

II. Related work

Educational mobile applications are an important part in the development of different areas, so they are a good support for adults and young people in the learning process. Within the topic of sexual health in [6] the authors indicate that the use of these applications generates a connection with young people, answering delicate questions that are generally difficult to deal with. Moreover, they provide a framework for identifying attitudes, norms, and perceived control that leads the person to intention and subsequently actions. Thus, these applications help to increase awareness of different factors that could prevent a person's capacity to act in a certain way. This also helps adults and teenagers to understand the points that impulse a human being's behavior to generate a positive change, thereby providing a process of sexual health development while adopting good practices of hygiene.

An example of this type of application is described in [7], where a game is presented, it is developed in Unity 5 and it uses educational tools such as stories, scenes, characters, settings, environments, badges, etc., to develop learning content about sexual health. The approach behind this proposal is interactive and aims to instill practices that repress reckless sexual behavior using game-based teaching methods and gamification to access the content.

Similarly, in [8] the authors developed an adventure game called "PlayForward", which is focused on roles within two dimensions whose context unfolds in decision-making inside a social environment whose results have a short- and long-term impact. The main objective of the game is to acquire and practice skills in order to reduce risky behaviors, foster healthier knowledge, attitudes and intentions regarding sexuality, in a variety of behaviors and risks such as HIV prevention. The game consists of 12 story-based challenges that are part of the general narratives.

Another application focused on sexual health is the one presented in [9] called "The Source", which shows the story of Adia, a 17-year-old Afro-American girl who finds a letter from her father who abandoned her and her family for unknown reasons. It presents a series of challenges, puzzles, board games, digital media tasks and treasure hunts that Adia tries to solve with a group of teenagers. In each activity, the content described per week is about sexual health.

In [10] the authors describe a ludic approach with immersive games that contain engaging narratives, fantasy elements, and interactive features that provide meaningful comments and rewards which are related to positive outcomes. All this with a sexual health perspective that goes from less mature to a more mature content as the game progresses.

In the proposal described in [11], the "Sex Talk" application is presented, which is an agenda for inclusive and permissive sex-education that presents a series of activities designed to develop a discussion about sex and sexuality. "Sex Talk" is a multiplayer game that was developed for iOS and designed to be played in groups. A point-to-point network is used here, through Wi-Fi or Bluetooth, in which players must turn their phone upside down until one of them vibrates, then the person has to turn it over and carry out the designated activity. This process is repeated simultaneously until all the players have completed all the tasks.

Otherwise, in [12] an application called "STD PONG" is presented, which is an interactive arcade game that takes place in a small town (clan) that has been controlled by an evil king and his henchmen, in which a chosen hero must fight with them to free his clan. For every defeated henchman, the player gains a scroll that contains information about sexual health. The general objective of the game is to acquire the nine scrolls of knowledge in order to finally free the clan from the evil king and his henchmen.

III. General system architecture

As can be seen in Figure 1, the educational game uses an approach to teach SHH concepts to young people from an interactive perspective based on cognitive, tactile, auditory and visual stimuli. The teachers and the teenagers can interact with the game through a tablet or smartphone (**user interface layer**). In the **services layer**, the game provides functionalities to manage the users' profiles (for both teenagers and teachers), the game also offers the possibility to monitor the learning progress of the teenagers and generates reports about this.

On the other hand, the **knowledge layer** has the duty to determine what educational strategies should be applied to the teenager according to their prior knowledge, gender and level of disability. For this, the expert system is based on the argumentation of rules to establish which application activities should be reinforced in order to promote learning towards a healthy sexual life. In the following subsection, we will describe the information (inference engine) that is required to arrive at a reasoning process that will later indicate

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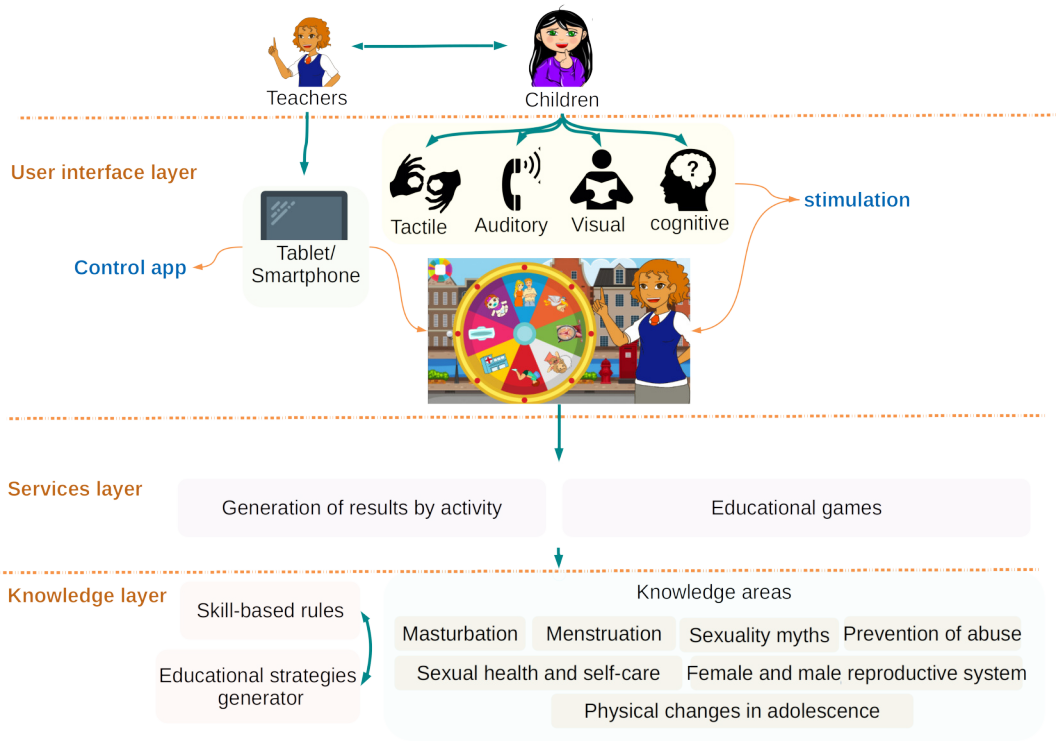


Fig. 1. Main elements of the proposed system.

what activities must be carried out to facilitate the teenagers learning. The areas of knowledge that are considered within the game are the following:

- **Know yourself:** this area of knowledge addresses the relevant aspects of masturbation and provides guidelines for executing this action in an appropriate and safe way. This section also promotes the importance of knowing your own body from the point of view of respect and acceptance.
- **Love and take care of yourself:** this section focuses on aspects that allow the user to maintain an adequate sexual health and hygiene, since it is vitally important that teenagers with intellectual disabilities learn to take care of themselves and have good hygiene habits to guarantee their sexual health.
- **How does your body change?:** this section consists of educational content about the physical and emotional changes that boys and girls experience until reaching adolescence, it also contains educational aspects about the process of falling in love. Moreover, this section focuses on the educational elements about the changes experienced by the male and female reproductive systems during the stages mentioned above.
- **Learn to say no!:** this section contains guidelines to avoid sexual abuse of teenagers with intellectual disabilities, since they are part of the vulnerable population and particularly sensitive to this type of abuse, especially teenager girls. In addition, it focuses on explaining the different body regions that should not be touched by anyone and how one should react to such situations.

A. *Expert system*

The inference engine employs four templates representing the following concepts:

- Teenager profile ($P = \{dl, a, g\}$): this structure stores the information that will be used to identify the specific subject. dl represents the level of disability, a the age and g the gender.
- Strategies (e): this structure is used to describe the strategies that will be used to support each activity.
- Educational activity (ac): it contains guides for teachers to assist them in the teenagers' learning process. One activity can be formed by one or more strategies. Currently, the system has 7 defined strategies that can be expanded as more material is created.
- Skills (S): this structure is used to determine the result of the activity. In this case, there are thirteen activities that are symbolically evaluated as "M," "B" and "N". The symbol "M" indicates that the activity has not been carried out correctly, "B" indicates that the objective has been achieved and "N" shows which activities are not applicable.
- Plan (pl): this structure is formed by the description of the strategies that will be applied according to the identifier of the person and consistent with what is illustrated in Eq. 1:

$$E = \bigcup_{i=1}^{N} \phi(s_i, P) \quad \text{s.t. } s_i \in S \ \wedge g \in P \ \wedge \ SC(s_i) = SC(g) \tag{1}$$

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Where SC represents a semantic function that checks if the concepts of SHH correspond to the gender of the teenager (por example, "How to put on the sanitary pad" is a topic that must be addressed for women). phi is the set of rules that the expert system implements and it receives the skills (s_i) as parameters, from a total of N skills and the profile of the teenager (P). The result is a set of strategies (E) that have to be applied by the teacher and that is obtained from the union of each strategy that the function returns $\phi(s_i, P)$.

Finally, it should be noted that in general, the system consists of four templates and twelve rules that will help us to determine which strategies will be implemented in order to obtain a correct learning. The rules are made up of eleven assumptions, evaluated according to the skills, thus generating the strategies that go from the informative reinforcement in the knowledge area, to the recursive application of the activity. Figure 2 shows an example of two templates used to represent the person, the intervention plan (upper part) and the work plan that the expert system returns based on a given input.

```
(deftemplate person
    (slot idperson (type NUMBER)
        (default 0)
    )
    (slot names (type STRING))
    (slot surnames (type STRING))
    (slot age (type NUMBER)(default 0))
    (slot gender (type SYMBOL)(allowed-symbols F M)(default F))
)

(deftemplate plan
(slot idplan (type NUMBER)(default 0))
(slot description (type STRING))
(slot idperson (type NUMBER)(default 0))
)

Intervention Plan.
===== INPUT =====
Profile = {Gender = Male, Age = 13, S = {B,B,B,M,N,M,N,M,B,M,M,B,B},
Intellectual Disability = Mild }

===== OUTPUT =====
E2. Informative explanation of the female reproductive system.
E4. Informative explanation to put on the sanitary pad.
E5. Informative explanation about the conception process.
```

Fig. 2. Example of two templates used in the expert system (upper part) and the result of the plan returned by the system (bottom).

B. Educational game

The SHH educational content of the game is supported through images that have been created with facial and body features similar to those of teenagers with intellectual disabilities, between 12 and 17 years old, in order to make them feel identified (Figure 3). Furthermore, these images have striking colors and contrasts to facilitate their visualization in case of comorbidities in relation to some type of visual deficit. Finally, it also has a voice reader to ease access to information for teenagers with visual deficits or for teenagers who have difficulty reading.

IV. Experiments and Results

In order to determine if this proposal can be used as an educational support tool, a survey was applied to 30 teachers who work in two special education institutions of Cuenca, Ecuador: Azuay's Institute of Cerebral Palsy (IPCA acronym in Spanish) and ADINEA. Professionals from the areas of pedagogy, psychology, speech therapy and physiotherapy, aged between 20 and 65 years, were surveyed.

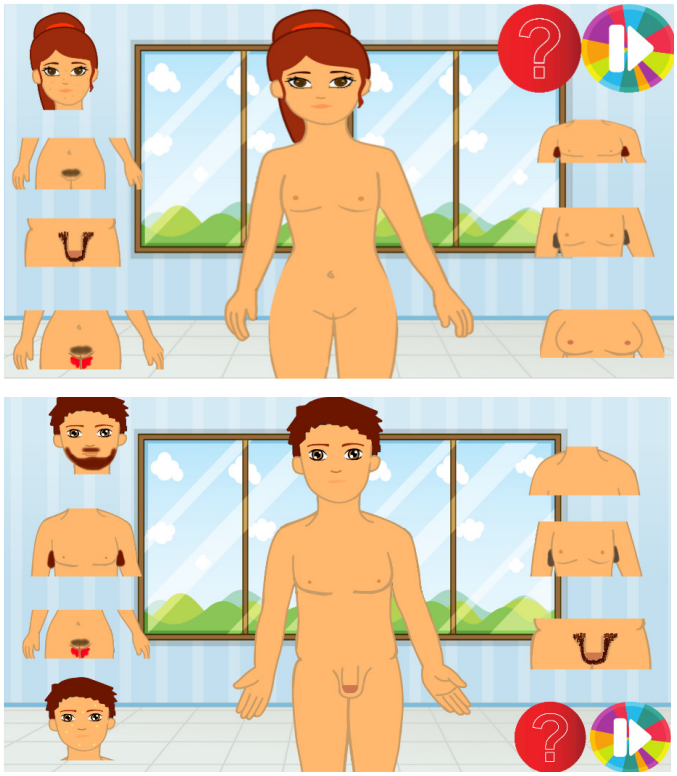


Fig. 3. Two screenshots from the game How does your body change?, the down image shows the man's body and the upper image shows the woman's body. Teenagers must identify the physical changes of each gender.

The analyzed data from the survey applied to the teachers indicated above were previously validated using the Cronbach's Alpha Test. The quotient of this test was 0.84, therefore, this index reflects an internal coherence between the items (questions) of the survey that was carried out.

It should be noted that the survey's structure was based on the criteria of various experts from different areas such as: special educators, software engineers, doctors, sexologists, anthropologists and statisticians; meaning that the survey was built using a multidisciplinary team.

The sample for the survey validation, as well as for the data analysis, was of a purposeful nature, since the persons selected to answer the survey were teachers from IPCA and ADINEA who could participate voluntarily in order to determine the perception of use from the teenagers with mild and moderate intellectual disabilities between 12 and 17 years old. Prior to the application of the survey, we proceeded to demonstrate to the aforementioned professionals the content of the Sexual Health and Self-care guide and the interactive games related to this guide, which are: How does your body change? and Learn to say no! Likewise, respondents were also instructed on how to complete the survey.

The most relevant results regarding descriptive statistical

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analysis (measures of central tendency) applied to the data obtained from the aforementioned survey are presented below. In Figure 4, it is observed that the majority of respondents are teachers (17 out of 30 respondents are from IPCA), who belong to the female gender (11) and with ages between 27 and 54 years old, followed by psychologists of the same gender, between 20 and 35 years old and the majority are from IPCA.

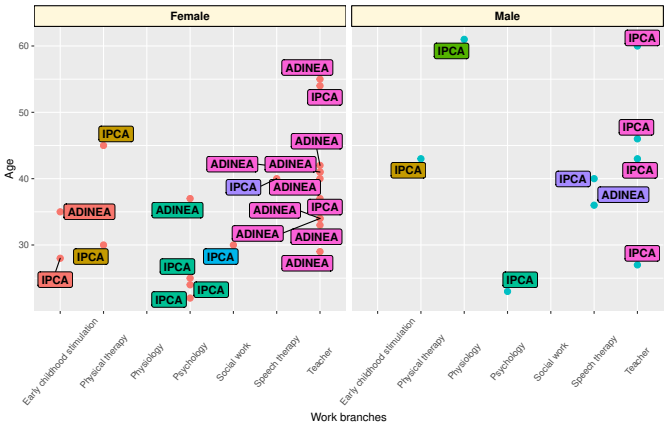


Fig. 4. Main demographic aspects of the experts who participated in the survey.

It should be taken into account that 21 of the 30 respondents are women from both institutions, IPCA and ADINEA, while only 9 are men. The majority of respondents are between the ages of 31 and 40, and 18 of the 30 respondents belong to IPCA. Moreover, Figure 5 shows that both female and male teachers from IPCA and ADINEA "totally agree" and "agree" with the language of the educational content used in the educational games of sexual health and self-care for students with intellectual disabilities between 12 and 17 years old. The perception of these teachers of specialized education prevails in different work areas such as: Early childhood stimulation, Physical therapy, Physiology, Psychology, Social work, Speech therapy, Teacher.



Fig. 5. Perceptions of the surveyed experts regarding the content and language that the educational game uses to teach SHH to teenagers with intellectual disabilities.

With the results and previous analysis, it can be said that the applied survey reflects the importance of the proposed tool to improve access to sexual health and self-care education for teenagers with intellectual disabilities with ages between 12 and 17 years old; but, above all, to improve the processes of educational inclusion and health for vulnerable populations such as the one described, since it requires priority attention.

V. Conclusions and Future Work

The Sexual Health and Self-care App, ITaCaS, is an inclusive educational tool that will allow access to content about sexual health and abuse-prevention to teenagers with intellectual disabilities between 12 and 17 years old, being a support tool for teaching either at classroom or at home that adjusts to the needs of the aforementioned population. This educational game has an adequate language, explicit images and content according to the age of the students and activities that aim to generate significant learning. It is important to mention that the application is a contextualized technological tool that seeks to answer a contextual social problem. Additionally, in the present investigation, it was evident that most of the teenagers who participated in the study have access to Internet and to mobile telephony, which are resources that facilitate the availability of technological educational tools that promote the learning of the students. It should be mention that among the limitations of the present study, there was limited openness of some parents, mainly because of the fact that they expressed feelings of amazement, fear and rejection when they learned that their children and/or those represented would interact with the app ITaCaS about educational topics of sexual health and self-care through educational games; and, they refused to allow their children and/or those represented to access the platform.

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